SANGUINE

OS11 - System Architecture

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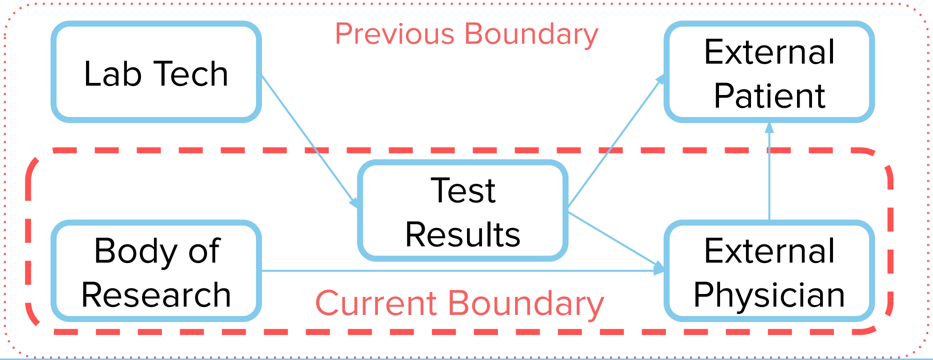
Q1: Refined SPS

To improve external patient care outcomes at scale
By augmenting physician-facing test results with relevant research
Using an agentic AI clinical librarian.

We chose this amount of breadth to clarify the scope. In previous versions (see appendix 1), we discussed both the scope and the architectural decisions. Narrowing our By and Using let us focus on the architectural decisions, while keeping our To focus on external patients. We could expand to internal patients someday, but for now, some requirements from OS9+10 are now out of scope, such as F.2. This is **still solution-neutral** because there are still many architectural decisions to be made, as Q2 shows. This diagram shows elements of the data flow we had previously considered, and which ones are in the current boundary implied by this SPS.



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Reference: OS9+10 for Traceability

Later answers trace back to our stakeholder needs and requirements from OS9+10. Those are summarized here for reference. All material on this slide is repeated from OS9+10, with minor updates for feedback. **This slide should not be graded for OS11.**

Primary Stakeholders and Needs

1. Patients (Primary beneficial):
   - 1.A. Personalized health status explanation
   - 1.B. Empowered actions
   - 1.C. Reassurance
2. External Physicians (Primary charitable):
   - 2.A. Info on medical specialties
   - 2.B. Standard workflows
3. Mayo Clinic Physicians (Primary charitable):
   - 3.A. Bolstering existing collaborative care
4. Mayo Clinic Leadership (Primary beneficial):
   - 4.A. Confirm Viability
5. Mayo Clinic Legal (Secondary problem):
   - 5.A. Mitigation of liability

Non-User/Operator Needs

6. Regulators, US and beyond (Secondary problem):
   - 6.A. Compliance Transparency
7. Insurance providers
   - 7.A. Reasonable Cost
8. Software Developers
   - 8.A. Defined interfaces and feedback loops

Functional Requirements

#	Name
F.1	Collection of clinical perspectives
F.2	Patient persona identification
F.3	Patient result personalization
F.4	Clinical reference retrieval - cases
F.5	Clinical reference retrieval - sources
F.6	Result interpretation support - ranges
F.7	Result interpretation support - trends
F.8	Patient-facing result presentation
F.9	Workflow integration - logins
F.10	Workflow integration - clicks
F.11	Alert management
F.12	Transparency of logic
F.13	Support for follow-up clinical conversations
F.14	Historic result context support
F.15	Empowered patient actions
F.16	Result accuracy transparency
F.17	Clinical librarian support
F.18	Compliance audit logging
F.19	Developer interface documentation

Non-functional Requirements

#	Requirement name
NF.1	Scalable interpretation throughput
NF.2	Multi-perspective interpretation
NF.3	Patient-level clarity
NF.4	Empathetic communication
NF.5	Specialists-based perspectives
NF.6	Mayo Clinic model alignment
NF. 7	Regulatory compliance
NF. 8	Result availability notification
NF.9	Cost efficiency

Q2: Candidate Architectural Decisions

ID#	Decision	Description	Options	Scope	Traced Reqs.	Commentary
D1	Compute runtime	Where to run our compute and AI inference.	[Edge, On-prem data center, Single-cloud *(GCP)*, Multi-cloud]	Form	F.18, F.19, NF.7, NF.9	Determines many aspects of cost, capabilities, and security. Mayo Clinic has a strategic partnership with Google Cloud, but we should still consider alternatives.
D2	Model Origin	What type of baseline model to build upon.	[Build-our-own, Open-source *(e.g., Llama)*, Closed-source *(e.g., Gemini, Claude)]*]	Form	F.12, F.19, NF.1, NF.9	We choose model origin instead of family (e.g., Gemini vs Claude as options) because within an origin, families are similar. We choose this instead of parameter count because origin is more connected to other decisions.
D3	Research paradigm	How to make the model know clinical background knowledge.	[RAG, Fine-tuning, Rely on pre-training]	Form - Function Mapping	F.4, F.5, F.14, F.17, NF.5, NF.9	Ways to augment clinical knowledge can be powerful, but restrictive and expensive. It also illustrates how “do nothing extra” can be an architectural choice.
D4	Communication channels	How to communicate non-result info to external physicians.	[EMR only, Web UI only, App only, Web+App, Email + EMR, App+Fax]	Form - Function Mapping	F.3, F.8	This is about the form-function mapping, not just form, because different channels (forms) could be deliver (be the instrument of) different communications (functions).
D5	Number of AI roles per test result	How many AI agents to divide the functions among.	[0 (non-agentic AI), 1, 2, 3-5, 6-10]	Form - Function Mapping	F.1, F.7, NF.2	Will determine which agents (forms) handle which capabilities (functions). If we have more agents, that form choice would then require new supporting functions, like coordination overhead.
D6	Data sources	Where the clinical librarian will be able to find data sources to share with physicians.	[Mayo internal only, 1-2 external sources, 3-5 external sources, Open Internet]	Function	F.1, F.4, F.5, F.12, F.17, NF.9	This determines the scope of the function of librarian research. More sources allow more breadth, but would demand more supporting functions, such as security support if we allow the open internet.
D7	Coordination structure	How the AI roles collaborate with each other to fulfill tasks.	[Agents as Tools, Swarm, Agent Graph / Workflow, No coordination]	Form	F.1, F.7, NF.2	Will determine how the agents will work together to achieve objectives[7].
D8	Clinical system Integration	How the system interacts with external clinical information systems	[MCP-exposed tools, API tools]	Form - Function Mapping	F.11, F.13, F.15, F.17, F.18, NF. 5, NF.6	Will determine how agents will retrieve external clinical information.
D9	Clinical validation and release control	How the system determines the interpretive data can be exposed to patient or physicians.	[Rule-based condition, Human-in-the loop, Tiered release]	Function	F.11, F.15, F.16, F.18, NF.6, NF.7	This defines when and to whom the interpretative information are released.
D10	Private information handling	How model handles to AI models and agents protected health information (PHI).	[No private exposure, Scoped PHI, No patient-identifiable PHI]	Function	F.9, F.16, F.18, F.19, NF.7, NF.9	This defines how sensitive patient data can be used by models and agents.

Q3: Metrics

#	Metric (Measurable)	Definition / Measurement	Primary Stakeholders	Stakeholder Needs	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
M1	**Patient Comprehension Score**	% of patients correctly answering 3–5 post-result comprehension questions (survey or in-app quiz)	Patients	1.A, 1.C	N	N	H	L	M	M	N	N	H	N
M2	**Patient Anxiety Delta**	Change in validated anxiety score (e.g., short STAI) pre- vs post-result explanation	Patients	1.C	N	N	L	L	N	L	N	N	H	N
M3	**Action Enablement Rate**	% of patients who can correctly identify next steps (non-advisory) after viewing results	Patients	1.B	N	N	L	L	M	M	N	N	H	N
M4	**Physician Time-to-Insight**	Median time (minutes) for physician to reach an interpretable understanding of result	External Physicians	2.A 2.B	H	L	H	M	L	N	L	M	L	N
M5	**Reference Utilization Rate**	% of reports where Mayo-curated references are accessed by clinicians	External & Mayo Physicians	2.A, 3.A	L	L	H	N	H	H	N	L	H	N
M6	**Clinical Accuracy / Concordance**	Agreement rate between AI output and Mayo specialist review (gold standard)	Patients, Physicians, Regulators	1.C, 3.A, 6.A	N	H	H	N	L	N	L	N	H	N
M7	**Compliance Evidence Coverage**	% of outputs with auditable provenance, citations, and versioned documentation	Legal, Regulators	5.A, 6.A	N	L	H	N	L	H	N	L	H	H
M8	**Cost per Interpreted Result**	Fully loaded cost ($) per explained result	Leadership, Insurance	4.A, 7.A	M	H	M	L	M	L	M	L	L	N
				Sum of Sensitivity Scores	16	23	57	9	28	32	7	8	62	10

No Sensitivity (0) | Less Sensitive (1) | Moderate Sensitive (5) | Highly. Sensitive (10)

The metrics come directly from the stakeholder analysis in IAP. Each metric measures if the main stakeholder needs (patients, physicians, leadership, legal, regulators) are met.
More important stakeholders are measured with more important metrics like accuracy and understanding.

The two most important metrics are:
1. Clinical Accuracy / Concordance (M6)
This metric is most important because incorrect interpretation can harm patients and create legal and regulatory risk.
2. Physician Time-to-Insight (M4)
This metric is important because, since as a clinical librarian, informing the physician is our main value-add.

We chose these two metrics because they were the most important to our end users, but noticed how these metrics are also two of the most sensitive!

D3 – Most sensitive to both M6 and M4: Strongly affects **clinical accuracy (M6)** because the knowledge-handling approach directly determines correctness and interpretability, and also impacts **physician time-to-insight (M4)** depending on reasoning structure and explainability.
D2 – Highly sensitive to M6; limited impact on M4: Different model architectures and training regimes directly influence reliability and medical safety, significantly affecting **clinical accuracy (M6)**, with comparatively smaller impact on interpretation time.
D1 – Primarily sensitive to M4; indirect impact on M6: System performance, latency, and deployment reliability affect **physician time-to-insight (M4)**. **Clinical accuracy (M6)** may be indirectly influenced if infrastructure instability affects output consistency.
D9 – Highly sensitive to M6; minimal impact on M4: Validation and release controls determine whether incorrect outputs are detected before deployment, directly safeguarding **clinical accuracy (M6)**.
D4 – Less sensitive: Primarily affects presentation and delivery format rather than **clinical correctness (M6)** or core interpretation time (M4).
D5 – Less sensitive: Influences workflow design but has limited direct impact on **clinical accuracy (M6)** or **physician time-to-insight (M4)**.
D6 – Less sensitive relative to D2 and D3: While data quality can influence accuracy, its marginal impact on **M6** is moderated by validation controls (D9).
D7, D8, D10 – Less sensitive: Mainly affect governance, compliance, and system integration rather than direct clinical correctness or physician interpretation speed.

Q4: Coupling

	D1 Compute runtime	D2 Model origin	D3 Research paradigm	D4 Communication channels	D5 Number of AI roles per test	D6 Data sources	D7 Coordination structure	D8 Clinical system integration	D9 Clinical validation and release control	D10 Private information handling	Total
D1 Compute runtime		X								X	3
D2 Model origin	X		X		X					X	4
D3 Research paradigm		X				X			X		3
D4 Communication channels								X	X		2
D5 Number of AI roles per test		X					X		X		2
D6 Data sources			X					X		X	3
D7 Coordination structure					X						1
D8 Clinical system integration				X		X				X	3
D9 Clinical validation and release control			X	X	X						3
D10 Private information handling	X	X				X		X			4
Total (# Related)	2	3	3	2	3	3	1	3	3	4	

Identified Clusters		D2 Model origin	D1 Compute runtime	D3 Research paradigm	D10 Private information handling	D6 Data sources	D8 Clinical system integration	D5 Number of AI roles per test	D9 Clinical validation and release control	D4 Communication channels	D7 Coordination structure	Total
Interpretation Infrastructure	D2 Model origin		X	X	X			X				4
	D1 Compute runtime	X			X							2
	D3 Research paradigm	X				X			X			3
Clinical Data Access and Integration	D10 Private information handling	X	X			X	X					4
	D6 Data sources			X	X		X					3
	D8 Clinical system integration				X	X				X		3
Interpretation Governance and Orchestration	D5 Number of AI roles per test	X							X		X	3
	D9 Clinical validation and release control			X				X		X		3
	D4 Communication channels						X		X			2
	D7 Coordination structure							X				1
	Total (# Related)	4	2	3	4	3	3	3	3	2	1	

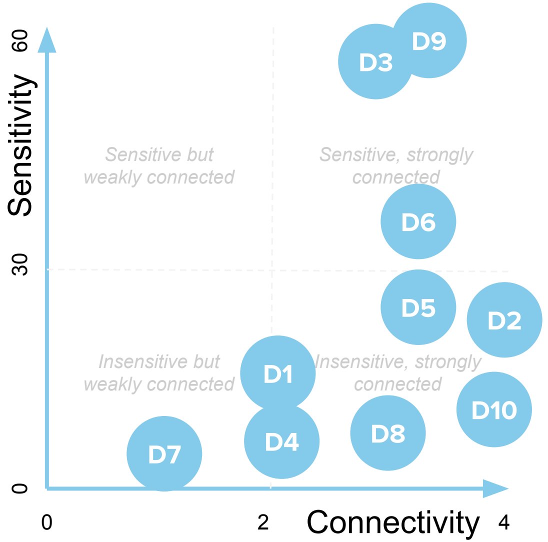
We constructed a DSM to represent the coupling relationships among all 10 decisions (D1-D10), where a dependency indicates that a change in one decision would require revisiting another. To identify groups of decisions that needs to be revisited together, we implemented the Louvain community detection method[1]. This method, when applied to a DSM, groups decisions with more interdependencies than other decisions, revealing sets of decisions that are highly coupled from an architectural perspective[2]. Using this approach, the technique identified three clusters in the DSM.

The first, "**Interpretation Infrastructure**", includes decisions on where the interpretation process runs and how clinical knowledge is incorporated.

The second, "**Clinical Data Access and Integration**", includes decisions about which clinical data can be accessed and how integrations with both internal and external systems are performed, observing privacy constraints.

The last cluster, "**Interpretation Governance and Orchestration**", includes decisions on how the interpretation layer is organized, including role decomposition, coordination mechanisms, validation, and release of results.

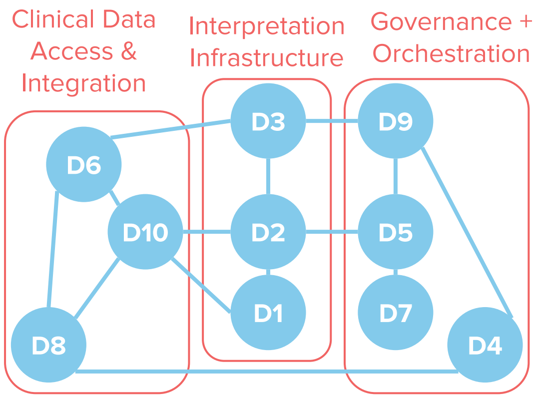
Q5. Sequencing Decisions



Ordering from most architectural to least:

- D9: Clinical validation and release control
- D3: Research paradigm
- D6: Data sources
- D5: #AI roles per test
- D10: Private information handling
- D2: Model origin
- D8: Clinical system integration
- D4: Comm. channels
- D1: Compute runtime
- D7: Coordination structure

The graph to the right shows decision connectivity, the same representation of the clustered DSM, but arranged by decision order from top to bottom. D3 and D9 are at the top and must be made together. D6 should be made next. Then, D10, D2, and D5 can be parallelized. After that, D1, D7, D4, and D8 can be parallelized because they aren't connected to each other. Our clusters from Q4 are apparent.



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Q6. Market Comparison

(Comparing architectures, not firms)

Current State: The current market has held onto stability for quite some time since the transition of medical test result delivery through internet or applications. This has been a consistent method of delivering results that include statistical “normal” ranges. However, it has grown stagnant, which is the catalyst for projects like this. The commoditization of LLMs is already starting to disrupt this stability; AI integration is commonplace.

Architectural Competition: Commoditized, static, “pdf summary” reporting is still dominant, but the market is at an inflection point. This is already seen on the patient side. Static delivery, like WebMD, is being replaced by LLM medical introspection[3]. While these competitive architectures are substitutes for the architecture we are proposing, they have shortcomings[4]. They lack clinical grounding and safeguards for patient data. While patients can give their own data to products like ChatGPT, this architecture lacks patient context, risk hallucination, and ultimately cannot meet the same level of care as a curated, guided, and compliant architecture. Our architecture is differentiated because it does not simply create a chatbot within a patient portal.

Architectural competitors include Labcorp’s Diagnostic Assistant[5] and Quest Diagnostics AI Companion [6] introduced to support physicians beyond static test outcomes. They offer a shift that Mayo Clinic is also seeking. While this shared competitive architecture seeks similar outcomes by addressing the same problems (needs) through similar methods (functions), the architectural difference is in the clinical librarian roles that agentic AI plays (forms). Mayo Clinic already has a competitive advantage through their collaborative panel of care. Our architectural interface mirrors this advantage to differentiate outcomes. The distinction between these offerings and ours is, their shared architecture assembles patient context and results, versus our architecture creating an interface layer with clinical interpretations.

Innovation versus existing architecture: The architectural approach under review is innovative. It disrupts the existing interface layer between test results, the patient context, and reporting structures. There are already some efforts to transition to architectures that echo what we are suggesting, but those do not create this synthesis interface layer to surface insights to physicians. Many of the components within the architecture are not novel and shared, but this interface layer is novel and breaks existing architectural approaches.

Appendix 1: SPS Evolution

Q1 shows our refined system problem statement. For context, the previous versions of our SPS are included here for context. All material on this slide is repeated from DC2, our project charter, and Q1. Changes are highlighted. **This slide should not be graded for OS11.**

Version	To	By	Using
DC2	improve external patient care outcomes at scale	clearly, compassionately, and collaboratively augmenting the information provided with Mayo test results	multi-specialized AI agents, such as multiple medical specialists
Project Charter	improve external patient care outcomes at scale	clearly, compassionately, and collaboratively augmenting the information provided with Mayo test results	**modern technology, such as AI agents**
OS11 Q1	improve external patient care outcomes at scale	**augmenting physician-facing test results with relevant research.**	**an agentic AI clinical librarian**

Appendix 2: Works Cited

[1] T. Aynaud, J.-L. Guillaume, R. Lambiotte, e col., “python-louvain: Louvain community detection in Python (documentation),” *python-louvain.readthedocs.io*. Acesso em: 08-fev-2026. [Online]. Disponível em: https://python-louvain.readthedocs.io/en/latest/

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